

THE BULL

Taurus



SIZE RANKING 17

BRIGHTEST STAR
Aldebaran (α) Tauri

GENITIVE Tauri

ABBREVIATION Tau

HIGHEST IN SKY AT 10 P.M.
December–JanuaryFULLY VISIBLE
88°N–58°S

Taurus is a large and prominent northern constellation of the zodiac, and it contains a wealth of objects including the Pleiades and Hyades star clusters (see p.291 and p.290) and M1, the Crab Nebula (see pp.270–271). Its stars represent the head and fore-quarters of a mythical Greek bull. The Hyades cluster is centered on the bull's face, while the constellation's brightest star, Aldebaran—Alpha (α) Tauri (see p.256)—is its glinting eye. Elnath (or Elnath)—Beta (β) Tauri—and Zeta (ζ) Tauri mark the tips of the bull's long horns. Each November, the Taurid meteors appear to radiate from a point south of the Pleiades.

SPECIFIC FEATURES

Aldebaran is a red giant whose color is clearly apparent to the naked eye. As with many red giants, it is slightly variable in brightness, but the amount is only about

THE CRAB NEBULA ☄

This supernova reveals the beauty of a massive star's violent death throes. Convoluted filaments of gas expand away from the site of the supernova explosion, which was seen from Earth in 1054 CE.

one-tenth of a magnitude either side of its average value of 0.85 and is barely noticeable. Although Aldebaran appears to be part of the Hyades cluster, it is 67 light-years away—less than half the cluster's distance—and is superimposed only by chance.

The main stars of the Hyades are arranged in a V-shape that is the width of over 10 diameters of a full moon. More than a dozen stars are visible with the unaided eye, and dozens more come into view through binoculars. At 150 light-years away, the Hyades is the closest major star cluster to Earth. On one arm of the Hyades' V-shape is a wide double star, Theta (θ) Tauri. At magnitude 3.4, the brighter of the pair, Theta-1 (θ^1), is also the brightest member of the Hyades. Another double star that is easy to spot is Sigma (σ) Tauri, which has two 5th-magnitude components, near Aldebaran. The apex of the Hyades cluster points toward Lambda (λ) Tauri, an eclipsing binary of the same type as Algol (in Perseus). It varies between magnitudes 3.4 and 3.9 in a cycle lasting just under 4 days.

An even brighter star cluster is the Pleiades, which hovers over the bull's shoulders. Although popularly known as the Seven Sisters, after a group of mythical Greek nymphs (see panel, opposite), the Pleiades in fact contains nine named stars: the seven sisters and their parents, Atlas and Pleione. The brightest member is Alcyone (see p.277), which is of magnitude 2.9 and lies near the center of the cluster. The Pleiades covers an area of sky three times the width of a full moon. On long-exposure photographs of the Pleiades, a surrounding haze is visible. This was once thought to be leftover gas and dust from the



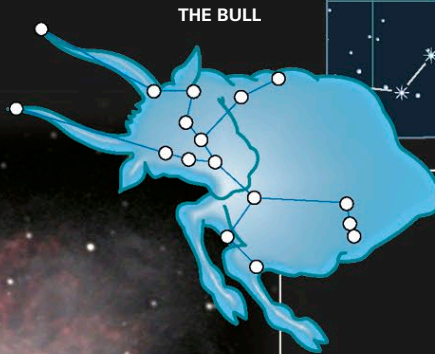
HYADES AND PLEIADES ☞

The Hyades (lower left) is the larger of these two dazzling star clusters; the Pleiades (upper right) is a tighter bunch that appears hazy at first glance—good viewing conditions are needed to see all nine named stars with the naked eye.

stars' formation, but it is now recognized as an unrelated cloud into which the cluster has drifted.

The first object on Charles Messier's list of cometlike objects (see p.73), M1 is the remains of a star that exploded as a supernova in 1054 CE. It was given its popular name, the Crab Nebula, by the Irish astronomer William Parsons (see p.315) in 1844, because he

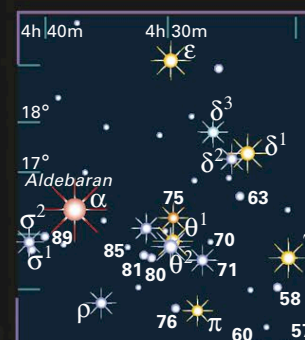
thought the filaments of gas that protruded from the supernova remnant resembled the legs of a crab. The Crab Nebula is found two diameters of a full moon away from Zeta Tauri. Through a small telescope, it appears as a faint elliptical glow several times larger than the disk of Jupiter. Large apertures are needed to make out the level of detail seen by Parsons.



THE PLEIADES



THE HYADES



MAGNITUDE KEY

